



COS80023 Big Data

Credit Task 6: Messaging with Kafka

Overview

Learn how to set up Kafka and what it does as a message broker.

Purpose

Experience how streaming data can be processed.

Task

Set up Kafka locally and stream messages from producer to consumer.

Time

This task is a Credit task and only mandatory if you would like to aim higher than a Pass grade. It should be submitted to Canvas for feedback. It should be discussed and signed off in tutorial 6 or 7.

This task should take no more than 2 hours to complete.

Resources

- Kafka from Canvas. This is an old version that works. You may use a newer version, but the tutorial may not be as helpful.
- Use this tutorial to set up an application using Kafka:
<https://kafka.apache.org/23/documentation/streams/quickstart>
- It is helpful and instructive if you go through this tutorial first (although you don't have to): <https://kafka.apache.org/quickstart>
- Any other online material.

Feedback

Demonstrate your solution to the tutorial instructor.

Next

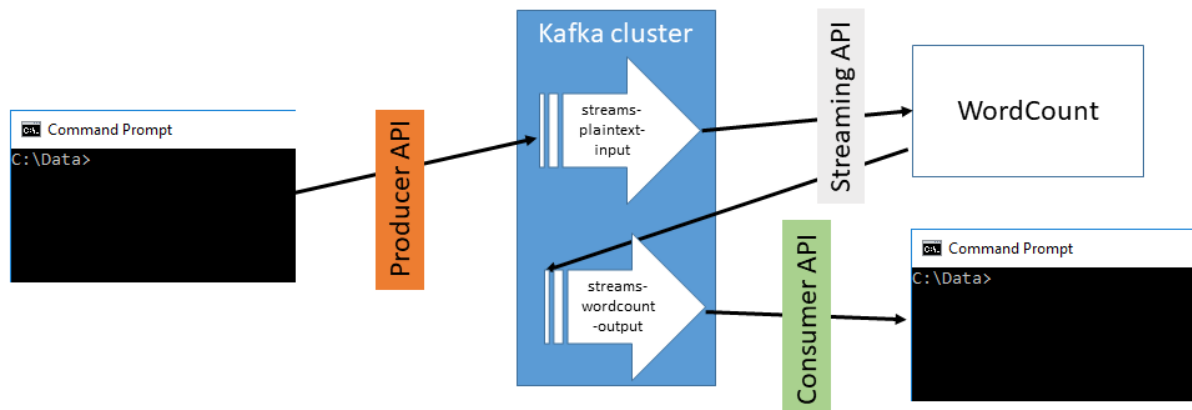
Get started on module 7.

Credit Task 6 — Submission Details and Assessment Criteria

For this task you only have to upload the screenshots of the producer and consumer consoles (terminals) when you have tested your setup. Your tutor will give online feedback and discuss the tasks with you in the lab when they are complete

Task 6

Set up a Kafka cluster on your local PC/Mac and configure it to use it with a wordcount application:



Overview of steps.

1. Launch a Zookeeper instance
2. Start a Kafka server.
3. Create topics `streams-plaintext-input` and `streams-wordcount-output`.
4. Start the Wordcount java module.
5. Start a producer that connects to `streams-plaintext-input`.
6. Start a consumer that connects to `streams-wordcount-output`. (You can also create a consumer that connects to `streams-plaintext-input` to listen to the raw data)
7. Type some text in the producer window and press Enter. Type some more and press Enter. Take a screenshot of your input and output windows and put into your document that you upload to Canvas.

Things to watch out for:

If you are using a Mac, the instructions in the tutorial apply. If you are using windows, you have to call `\bin\windows\<somescript>.bat` instead of `\bin\<somescript>.sh`

I had to change the `kafka-run-class.sh` and `kafka-run-class.bat` files, because the directory to the `kafka-streams-examples-2.3.0.jar` file was wrong. If you are downloading directly, you have to adjust this.

The `.properties` files in the config directory have the Linux file separator `/`. If you have downloaded the `.zip` file from Canvas, I have changed the log directory's separators to `\` in `server.properties`. Change back if you are using a Mac.

The exercise makes you open many command windows (or terminals). I suggest use one terminal to run all commands, because it preserves a history of commands and you can go back and change commands (using the 'up' button) without having to type or paste them again.

On a windows machine, you simply start the line with 'start', e.g.

```
start bin\windows\kafka-server-start.bat config\server.properties
```

On a Mac, you add `&` to the end to start running the program in the background:

```
bin/kafka-server.sh config/server.properties
```